Test Summary Report

Lost Grenadiers Navigation System

Version 1.0

Prepared by: Audrey Dickson, Kaden Langford, Andrew Swartz

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Contents

Test Summary Report 1
Purpose 3
Application Overview 3
Testing Scope 4
Metrics 5
Types of Testing Performed 7
Test Environment & Tools 8
Lessons Learned 9
Recommendations 9
Best Practices 9
Exit Criteria 10
Conclusion 10
Definitions, Acronyms, and Abbreviations 10

Purpose

This document summarizes the testing activities performed for the Lost Grenadiers Navigation System. It outlines the scope of testing, execution results, defect metrics, lessons learned, and final recommendations prior to production readiness.

The objective of this report is to formally document the quality status of the system after completion of planned testing.

Application Overview

The Lost Grenadiers Navigation System is a web-based campus navigation application developed for users to locate buildings, rooms, and generate routes both indoors and outdoors.

The system includes:

- Interactive map interface
- Real-time user geolocation
- Outdoor and indoor routing
- Visual floor overlay system
- Accessible route toggle
- Flask-based backend routing
- PostgreSQL database with pgRouting
- HTTPS secured deployment

Subsystems include:

- **Frontend UI:** JavaScript, map rendering
- **Routing Engine:** Flask API
- **Database Layer:** PostgreSQL, pgRouting
- **Security:** HTTPS, firewall, injection protection

Testing Scope

In Scope:

- **Frontend:**
  - Map rendering
  - User location loading
  - Search functionality
  - Valid/Invalid route generation
  - Accessibility filters
- **Flask API:**
  - Request validation
  - Route computation
  - Error handling
  - JSON responses
- **pgRouting/PostGIS:**
  - Shortest-path calculations
  - Data integrity
- **Security:**
  - TLS/SSL
  - SQLi prevention
  - Firewall/DMZ access control
- **Performance:**
  - Response times
  - Query execution
  - Render speed

Out of Scope:

- GPS signal quality or reception issues
- Accuracy of third parties (OpenStreetMap)
- User authentication (no login system exists)

Not Tested:

- Large scale concurrent user stress testing
- Full accessibility compliance audit

Metrics

- **Functional:**
  - Pass/fail of routing
  - Map rendering
  - Accessibility modes
  - Error handling
- **Security:**
  - TLS/SSL validation
  - API endpoint protection
  - SQLi prevention
- **Performance:**
  - Routing response times
  - Query execution
  - Concurrent user handling
- **Usability:**
  - Ease of navigation
  - Search accuracy
  - Clarity of route visuals
- **Defects:**
  - Logged with severity levels of critical, major, minor, cosmetic, and possible expected fix times for each

Test Cases Planned vs. Executed

Category	Planned	Executed
Frontend	6	6
Backend API	4	4
Security	3	3
Pure Backend	5	5
Total Test Cases:	**15**	**15**

All planned test cases were executed.

Test Case Results

Result	Count
Passed	15

Failed	0

All test cases passed after implementation fixes.

Defect Summary

Severity	Count	Status
Critical	0	N/A
Major	1	Closed
Minor	1	Closed
Cosmetic	0	N/A

Defects identified and resolved:

- HTTPS redirect initially not enforced.
- Minor overlay invocation behavior required improved testing approach.

All identified defects were corrected and retested successfully.

Observed Routing Response Time:

- Average: < .467ms
- Worst case tested: < 2.529ms

Types of Testing Performed

- **Functional Testing:**
  - Map rendering
  - Route calculations
  - API responses
  - Accessibility
- **Performance Testing:**
  - Load testing
  - Query benchmarks
  - Map responsiveness even with large amounts of nodes
- **Security Testing:**
  - TLS/SSL verification
  - SQLi attempts
  - API overloading
  - API malformed requests
- **Compatibility testing:**
  - Browser support for Chrome, Firefox, Safari, as well as OS compatibility
- **Usability Testing:**
  - UI clarity
  - Accessibility modes
  - Error messages in backend applications

Test Environment & Tools

- **Frontend URL:** https://lostgrenadiers.org
- **Hardware & Virtualization:** Proxmox VE hosting VM containing Flask API and PostgreSQL database
- **Software Platforms:**
  - **Backend:** Flask 2.X (Python 3.X), PostgreSQL 15, PostGIS, pgRouting
  - **Frontend:** Leaflet.js map interface, Cypress for frontend UI config testing
  - **Security:** Let's Encrypt TLS/SSL certificates, DMZ configurations
- **Network:**
  - Local LAN for penetration testing, HTTPS for all API calls, Proxmox network segmentation
- **Testing Tools Used:**
  - Browser DevTools
  - DBeaver
  - QGIS
  - Let's Encrypt TLS Certificate validation

Lessons Learned

- There are ~1,600 outdoor nodes and ~1,400 indoor nodes, both of which are split into separate tables. This node density does not degrade map rendering or routing speed, which means precomputation of common routes are not necessary. If we were to have double the nodes and double the edges, i.e. approximately double the campus size, then indexing would be advantageous performance-wise for the end user experience.
- Updating the graph in increments rather than in larger chunks reduces the odds of missed edges and impossible routing conditions.
- Automation tests for the API testing and load simulation greatly helps identify potential bottlenecks, though we did not experience any.
- Despite a frontend testing software, UI/UX ultimately must be tested manually for guaranteed user acceptance.

Recommendations

- Schedule frequent backups of the VM through Proxmox. This is a simple way to ensure that even a database corruption can be recoverable.
- Continuously run testing scripts to verify security of the application as well as common rendering. This could be integrated with GitHub to verify that all tests work before a PR is merged into the main branch.
- Auditing logs and noting defects could help in detection of various bugs or issues that may arise and are not noted in the current testing suite.

Best Practices

- Validate all user inputs with parameterization and sanitize for SQL queries.
- Ensure clear and informative error messages for the end user.
- Monitor system resource utilization during periods of higher load.
- Keep the environment configuration version-controlled with snapshots.
- Test usability across browsers and devices before each release.

Exit Criteria

- ✓ All core features pass predefined tests with no critical defects.
- ✓ TLS/SSL certificates are verified and API endpoints are secured.
- ✓ Routing queries complete with an acceptable response time.
- ✓ Database node and edge integrity confirmed.
- ✓ Documentation finalized with test cases.

Conclusion

The Lost Grenadiers Navigation System has been thoroughly tested across functional, performance, security, and usability concerns. All identified defects were documented and addressed based on severity to the integrity of the application as well as the ingress and egress network communications surrounding it. Based on the successful completion of all defined exit criteria, the testing team recommends approval for production deployment.

Definitions, Acronyms, and Abbreviations

- API – Application Programming Interface
- SQLi – SQL injection
- TLS/SSL - Transport Layer Security / Secure Socket Layer
- UI/UX - User Interface / User Experience
- Proxmox VE – Virtualization Environment